



AFLA BEGINNER GUIDE 2 OF 3

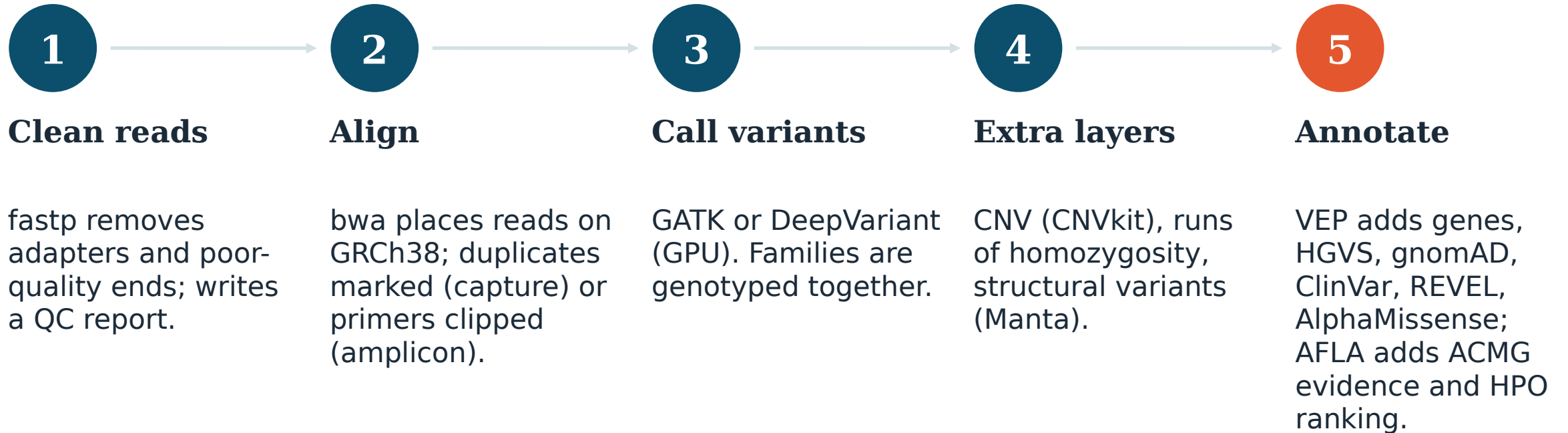
Germline analysis

From reads to a classified variant: QC, filtering, inheritance, ACMG/AMP evidence and the case report

Education and research use only · not for clinical diagnosis or patient management

What happens when you click Run

Every box can be switched on or off in the form



Setting up a case

Optional fields that make the report much smarter



Sample sheet (families)

CSV with sample, role, sex, family, hpo. Family members are called together, so de novo and compound heterozygous variants can be recognised.



HPO terms

The patient's features as HPO codes, e.g. HP:0001250 (seizure). Genes are ranked by how well their known phenotypes match.



Gene list / panel

Genes to focus on, or a PanelApp panel chosen inside the report: a "virtual panel" that avoids incidental findings.

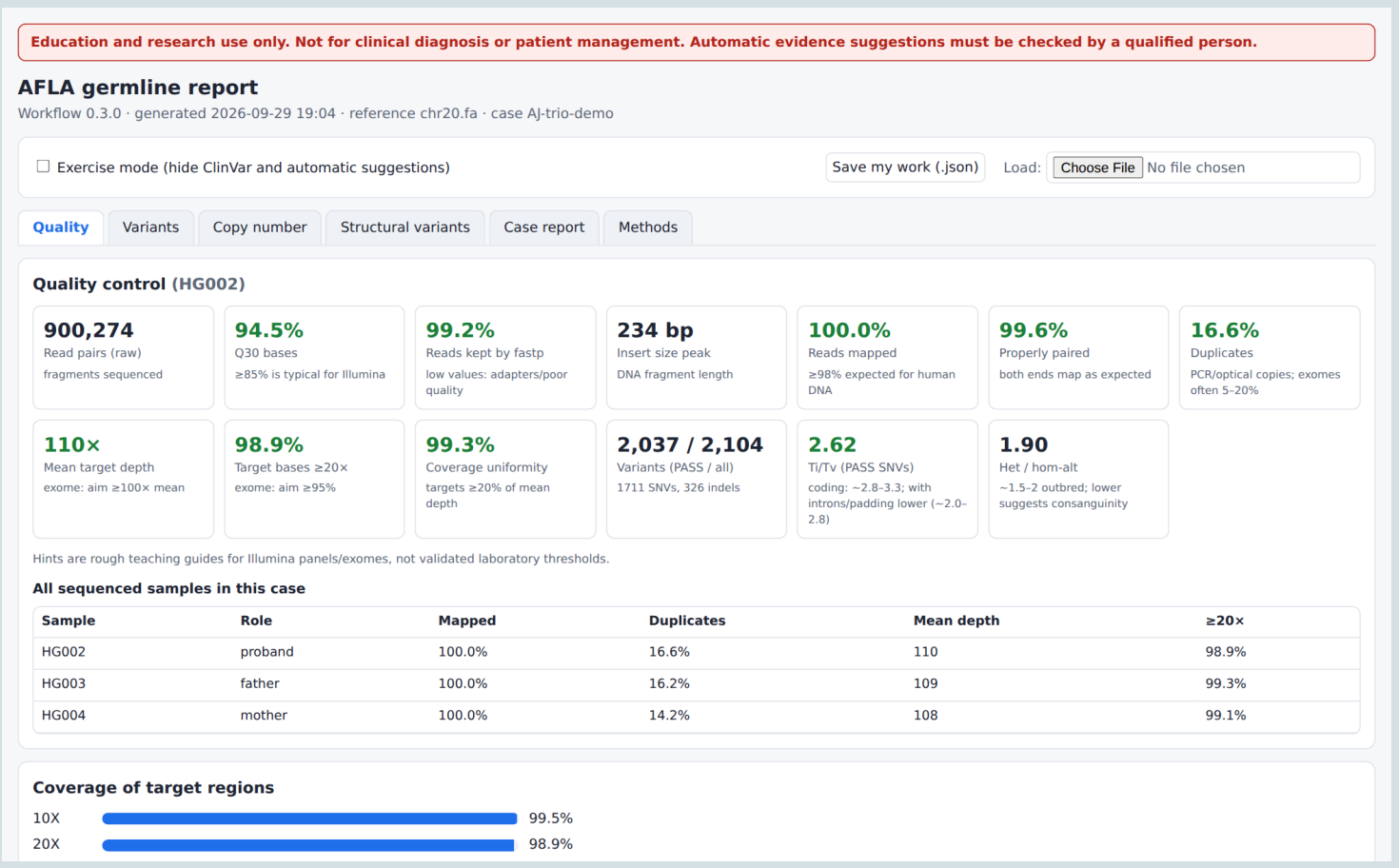


Case ID

A pseudonym printed on the report. Never use a patient's name.

Tab 1: Quality

Is the sequencing good enough to trust a negative result?



Tab 2: Variants

Presets do the first filtering for you

Education and research use only. Not for clinical diagnosis or patient management. Automatic evidence suggestions must be checked by a qualified person.

AFLA germline report

Workflow 0.3.0 · generated 2026-09-29 19:04 · reference chr20.fa · case AJ-trio-demo

☐ Exercise mode (hide ClinVar and automatic suggestions)

Save my work (.json)Load:

Choose FileNo file chosen

Quality

Variants

Copy number

Structural variants

Case report

Methods

Variant explorer

Prioritised (ACMG + phenotype)

Rare, protein-affecting

ClinVar pathogenic / likely pathogenic

De novo (trio)

Recessive (homozygous / compound het)

All variants

Quality

☒ PASS only

Min depth

0

Min VAF

0.2

Population frequency

Max gnomAD / max pop. AF

0.01

☒ keep variants with no known frequency

Predicted impact (VEP)

☒ HIGH☒ MODERATE☒ LOW☐ MODIFIER☒ + predicted splice effect

Genetics

Inheritance

any

ACMG (suggested)

VUS or worse

☐ phenotype-matched genes only (HPO)

ClinVar

☐ pathogenic / likely pathogenic only

Genes

or type genes: BRCA1, BRCA2, PALB2

Search

gene, HGVS, rsID...

22 of 2,104 variants shown

Click a column to sort, a row for details. ☆ adds a variant to the case report.

Export filtered rows (CSV)

☆	Gene	Variant	HGVSc	HGVSp	Consequence
☆	CTSA	chr20:45891598 CCTGCTG>C	c.105_110del	p.Leu36_Leu37del	inframe deletion
☆	GNAS	chr20:58840745 G>A	c.639G>A	p.Glu213=	synonymous variant
☆	VPS16	chr20:2860738 C>T	c.515-10C>T		splice polypyrimidine tract variant, intron variant
☆	GDF5	chr20:35434398 T>C	c.1017A>G	p.Lys339=	synonymous variant
☆	CEP250	chr20:35501830 CCT>C	c.382-10_382-9del		splice polypyrimidine tract variant, intron variant, NMD transcript var

Presets, filters and the variant table

Prioritised ACMG points + phenotype match + your gene list.

Rare, protein-affecting gnomAD ≤ 1%, HIGH/MODERATE impact.

ClinVar P/LP already reported as disease-causing.

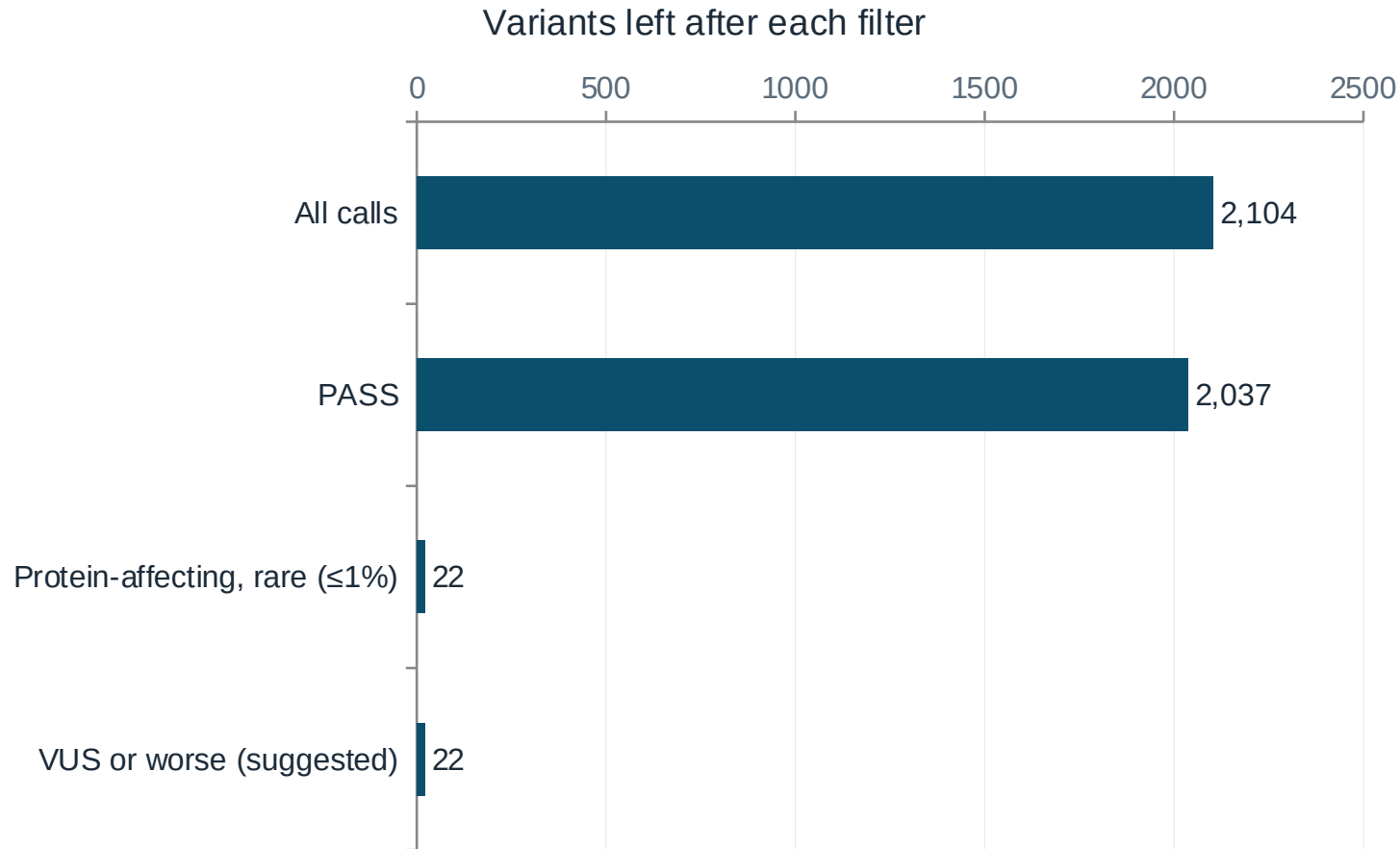
De novo / Recessive for trios.

Click a column to sort; a row to open details.

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Filtering is a funnel

Real numbers from one exome slice (HG002, chromosome 20)



- A whole exome has 20,000-50,000 variants; most are common and harmless.
- Filters remove what cannot explain a rare disease.
- What remains is reviewed one by one: that is interpretation.
- Too strict a filter can hide the answer: always know what you removed.

The variant card

Everything about one variant, plus the family

CTSA c.105_110del p.Leu36_Leu37del

☆ Add to the case report

Check current gnomAD / ClinVar online

Gene	CTSA (LOEUF 0.85, missense Z 1.65)	Transcript	ENST00000372484.7	Exon / intron	2/15	HGVSc	c.105_110del
HGVSp	p.Leu36_Leu37del	Consequence	inframe_deletion	Impact	MODERATE	Class	deletion
Genotype	1/0	Inheritance		VAF	0.88	Depth	152
GQ	99	QUAL	13170.2	Filter	PASS	gnomAD exomes AF - REVEL	-
gnomAD genomes AF - AlphaMissense	-	gnomAD (file) AF - SpliceAI (max Δ)	0.00366 -	Max population AF - SIFT		PolyPhen	
Known IDs		ClinVar	-	ClinVar review		ClinVar condition	
HPO match	0.42: Seizure ~ Seizure; Global developmental delay ~ Neurodevelopmental abnormality						

Genotypes in the family

Sample	Role	GT	Depth	GQ	VAF
HG002	proband	1/0	152	99	0.88
HG003	father	0/0	187	99	0.04
HG004	mother	0/0	192	99	0.02

ACMG/AMP evidence

Classification: Uncertain significance 2 points (Tavtigian et al. 2020: ≥10 pathogenic, 6-9 likely pathogenic, 0-5 VUS, -1 to -6 likely benign, ≤ -7 benign)

Reset to suggestions

☐ **PVS1** VeryStrong

Null variant (nonsense, frameshift, canonical ±1/2 splice, start loss) in a gene where loss of function causes disease

☐ **PS3** Strong

Well-established functional studies show a damaging effect

☐ **PM2** Supporting

Absent (or extremely rare) in population databases (ClinGen: supporting)

☐ **PM5** Moderate

Novel missense at a residue where a different missense change is pathogenic

☐ **PS1** Strong

Same amino-acid change as an established pathogenic variant

☐ **PS4** Strong

Prevalence in affected individuals significantly higher than in controls

☐ **PM3** Moderate

Recessive disorder: detected in trans with a pathogenic variant

☐ **PM6** Moderate

Assumed de novo (parentage not confirmed)

☐ **PS2** Strong

De novo, maternity and paternity confirmed, phenotype consistent

☐ **PM1** Moderate

Mutational hotspot or critical functional domain without benign variation

☒ **PM4** ModerateProtein length change (in-frame indel outside repeats, stop-loss)
suggested: in-frame deletion/insertion or stop-loss changes protein length (not applicable inside a repeat region)☐ **PP1** Supporting

Co-segregation with disease in multiple affected family members

- Gene, transcript** MANE Select transcript, exon, HGVS names.
- Frequency** gnomAD exomes/genomes, max population.
- Predictions** REVEL, AlphaMissense, SpliceAI.
- Family** genotype, depth and VAF of each member.
- Links** gnomAD, ClinVar, UCSC, ClinGen, OMIM, PubMed, PanelApp.
- Online check** current gnomAD/ClinVar with one click.

Detail card with family genotypes and links

Reading a trio

Inheritance labels AFLA adds automatically

Label	Child	Mother	Father	Think of
de novo	het ($\approx 50\%$)	ref	ref	dominant disorders, new mutations
possible de novo (low VAF)	het < 25%	ref	ref	usually artefact; sometimes mosaic
homozygous	hom	het	het	recessive disorders (consanguinity)
compound heterozygous	het + het	one	the other	recessive, two different variants
hemizygous (X-linked)	hom (male)	het	ref	X-linked recessive in boys
inherited from ...	het	het	ref	dominant if that parent is affected

In our GIAB test, all 5 "de novo-looking" calls had 8-17% of reads: artefacts. True constitutional de novo variants sit near 50%.

ACMG/AMP evidence panel

Tick the evidence; the class updates live

☐ PVS1 VeryStrong ▾ Null variant (nonsense, frameshift, canonical $\pm 1/2$ splice, start loss) in a gene where loss of function causes disease	☐ PS1 Strong ▾ Same amino-acid change as an established pathogenic variant	☐ PS2 Strong ▾ De novo, maternity and paternity confirmed, phenotype consistent
☐ PS3 Strong ▾ Well-established functional studies show a damaging effect	☐ PS4 Strong ▾ Prevalence in affected individuals significantly higher than in controls	☐ PM1 Moderate ▾ Mutational hotspot or critical functional domain without benign variation
☐ PM2 Supporting ▾ Absent (or extremely rare) in population databases (ClinGen: supporting)	☐ PM3 Moderate ▾ Recessive disorder: detected in trans with a pathogenic variant	☒ PM4 Moderate ▾ Protein length change (in-frame indel outside repeats, stop-loss) suggested: in-frame deletion/insertion or stop-loss changes protein length (not applicable inside a repeat region)
☐ PM5 Moderate ▾ Novel missense at a residue where a different missense change is pathogenic	☐ PM6 Moderate ▾ Assumed de novo (parentage not confirmed)	☐ PP1 Supporting ▾ Co-segregation with disease in multiple affected family members
☐ PP2 Supporting ▾ Missense in a gene with low benign missense variation where missense is a common mechanism	☐ PP3 Supporting ▾ Computational evidence supports a deleterious effect	☐ PP4 Supporting ▾ Phenotype or family history highly specific for a disease with a single genetic cause
☐ BA1 Allele frequency > 5% in a population database	☐ BS1 Strong ▾ Allele frequency greater than expected for the disorder	☐ BS2 Strong ▾ Observed in a healthy adult (homozygous for recessive, heterozygous for dominant fully penetrant disease)
☐ BS3 Strong ▾ Well-established functional studies show no damaging effect	☐ BS4 Strong ▾ Lack of segregation in affected family members	☐ BP1 Supporting ▾ Missense in a gene where only truncating variants cause disease
☐ BP2 Supporting ▾ Observed in trans with a pathogenic variant (dominant) or in cis (any)	☐ BP3 Supporting ▾ In-frame indel in a repetitive region without known function	☐ BP4 Supporting ▾ Computational evidence suggests no impact
☐ BP5 Supporting ▾ Variant found in a case with an alternative molecular cause	☐ BP7 Supporting ▾ Synonymous (or deep intronic) with no predicted splice impact and not highly conserved	

Interpretation note

Your reasoning, literature, segregation, phenotype fit...

Look up: [gnomAD](#) · [ClinVar](#) · [UCSC](#) · [ClinGen](#) · [OMIM](#) · [PubMed](#) · [PanelApp](#)

Criteria that need literature, functional data, segregation or phenotype (PS3, PS4, PP1, PP4, BS3, BS4, BP5...) are never ticked automatically. Annotations are evidence to weigh, not a classification.

Suggested from data: PVS1, PS1, PM2, PM4, PM5, PP3/BP4, BA1, BS1, BP7, PM6, PM3.

Never automatic PS3, PS4, PP1, PP4, BS3, BS4, BP5: need literature, segregation or phenotype.

Strength change it with the drop-down (e.g. PP3 strong for REVEL ≥ 0.932).

Note write your reasoning; star ☆ to report.

All 26 criteria; suggested ones pre-ticked with the reason

From evidence to a class

Bayesian points (Tavtigian et al. 2020)

8

Very strong (e.g. PVS1)

4

Strong (e.g. PS1, PP3 strong)

2

Moderate (e.g. PM4, PM5)

1

Supporting (e.g. PM2, PP3).
Benign criteria count negative

Total points	≥ 10	6 to 9	0 to 5	-1 to -6	≤ -7 (or BA1)
Class	Pathogenic	Likely pathogenic	Uncertain significance	Likely benign	Benign

Example: stop-gain in a loss-of-function intolerant gene (PVS1 = 8) absent from gnomAD (PM2 = 1) and de novo (PM6 = 2) → 11 points → Pathogenic.

Phenotype matching (HPO)

Which genes fit the patient?



You enter features

e.g. seizure (HP:0001250) +
global developmental delay
(HP:0001263).



AFLA compares

with each gene's known
features in the Human
Phenotype Ontology (semantic
similarity, 0-1).



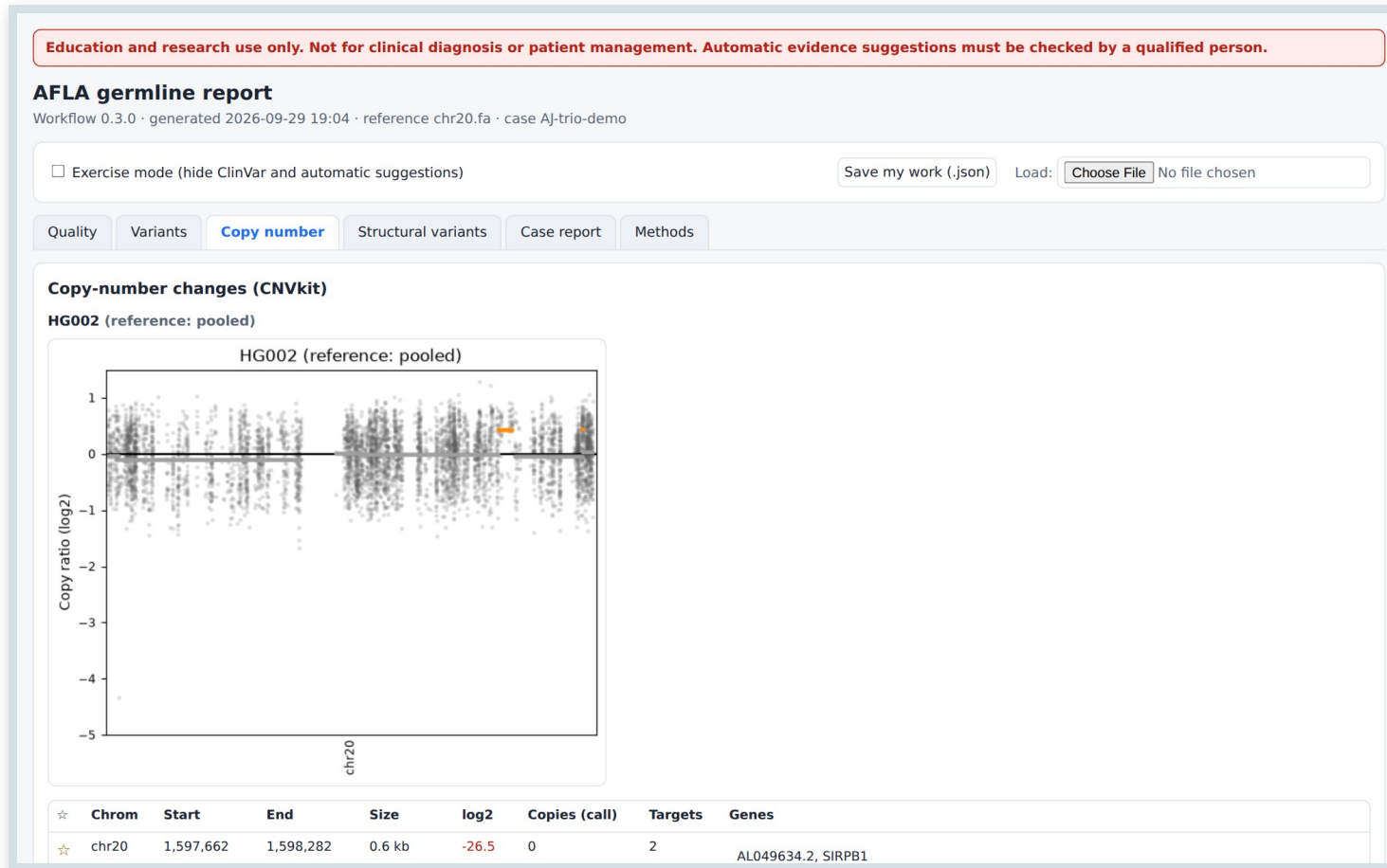
Genes rise

In our test, epilepsy genes
(SLC12A5, KCNB1, KCNQ2,
EEF1A2) ranked top on
chromosome 20.

- A high HPO score is support, not proof: the healthy GIAB sample has none of these diseases.
- Precise, specific terms help more than general ones ("infantile spasms" beats "seizure").

Copy number and homozygosity

Beyond single-letter changes



CNVkit plot and segments (reference: other samples of the run)

log2 ≈ -1 one copy lost; $\approx +0.58$ one copy gained.

Reference matters "pooled" (3+ samples, same kit) or your own; "flat" is weak.

Confirm CNVs by MLPA, array or qPCR.

ROH long homozygous stretches: consanguinity, and where recessive genes hide.

Exercise mode and the case report

Learn by doing, then compare

Education and research use only. Not for clinical diagnosis or patient management. Automatic evidence suggestions must be checked by a qualified person.

AFLA germline report

Workflow 0.3.0 · generated 2026-09-29 19:04 · reference chr20.fa · case AJ-trio-demo

☒ Exercise mode (hide ClinVar and automatic suggestions)

Save my work (.json)

Load:

Choose File

 No file chosen

Quality

Variants

Copy number

Structural variants

Case report

Methods

Variant explorer

Prioritised (ACMG + phenotype)

Rare, protein-affecting

ClinVar pathogenic / likely pathogenic

De novo (trio)

Recessive (homozygous / compound het)

All variants

Quality

Population frequency

Predicted impact (VEP)

Genetics

☒ PASS only

Min depth 0

Min VAF 0.2

Max gnomAD / max pop. AF 0.01

☒ keep variants with no known frequency

☒ HIGH

☒ MODERATE

☐ LOW

☐ MODIFIER

☒ + predicted splice effect

Inheritance any

ACMG (suggested) any

☐ phenotype-matched genes only (HPO)

Genes

Search

or type genes: BRCA1, BRCA2, PALB2

gene, HGVS, rsID...

18 of 2,104 variants shown

Click a column to sort, a row for details. ☆ adds a variant to the case report.

Export filtered rows (CSV)

☆	Gene	Variant	HGVSc	HGVSp	Consequence	GT	Inheritance
☆	CTSA	chr20:45891598 CCTGCTG>C	c.105_110del	p.Leu36_Leu37del	inframe deletion	1/0	
☆	GDF5	chr20:35434589 C>A	c.826G>T	p.Ala276Ser	missense variant	0/1	
☆	CEP250	chr20:35501852 T>C	c.389T>C	p.Leu130Pro	missense variant, NMD transcript variant	1 1	homozygous
☆	VSX1	chr20:25079507 G>C	c.432C>G	p.Asp144Glu	missense variant	0/1	inherited from father
☆	ADRA1D	chr20:4248728 G>T	c.230C>A	p.Ala77Glu	missense variant	0/1	inherited from mother

Exercise mode hides ClinVar and suggestions

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AFLA germline report

Workflow 0.3.0 · generated 2026-09-29 19:04 · reference chr20.fa · case AJ-trio-demo

☐ Exercise mode (hide ClinVar and automatic suggestions)

Save my work (.json)

Load:

Choose File

 No file chosen

Quality

Variants

Copy number

Structural variants

Case report

Methods

Print / save as PDF

Fields with dashed borders can be edited. Your changes are kept in this browser; use "Save my work" to keep a file.

EDUCATION / RESEARCH ONLY - NOT A CLINICAL REPORT

Genomic analysis report (teaching)

Case ID

AJ-trio-demo

Sample(s)

HG002, HG003, HG004

Test

chr20.targets.bed

Targeted sequencing:

Date

2026-09-29 19:04

Sex

male

Family

proband HG002; mother HG004; father HG003

Clinical information / indication

Phenotype (HPO): Seizure (HP:0001250); Global developmental delay (HP:0001263)

Summary

1 variant(s) reported below.

Sequence variants

Gene	Variant	Zygosity / inheritance	Classification (criteria)	Interpretation
CTSA	c.105_110del p.Leu36_Leu37del	1/0	Uncertain significance PM4	

Interpretation and recommendations

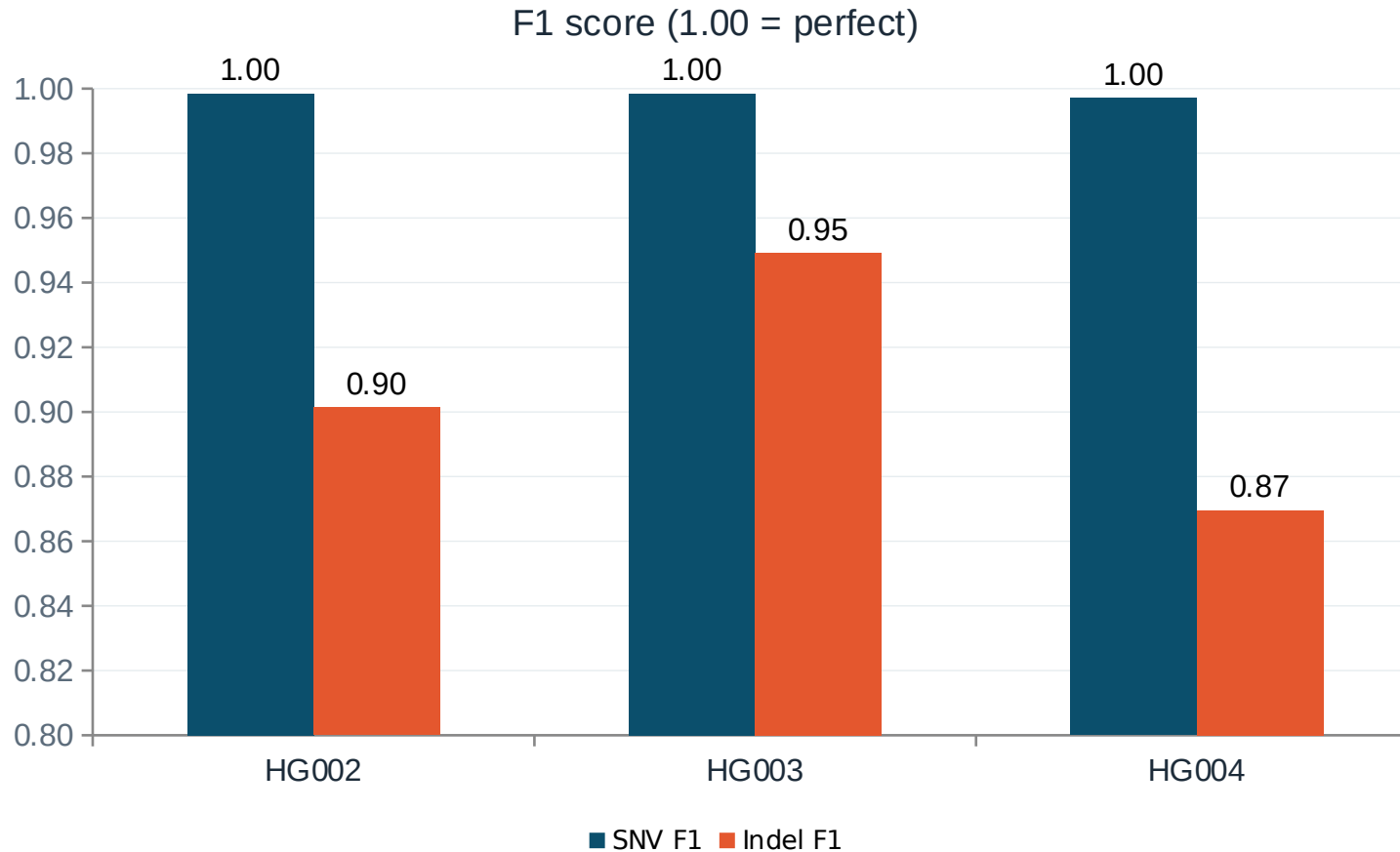
Printable case report built from your ☆ choices

- Students classify blind, then untick exercise mode to compare.
- "Save my work" keeps stars, criteria and notes in a small file for marking.

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How accurate is it?

Scored against Genome in a Bottle truth sets (chr20 exome targets, GATK)



- SNVs: > 99.7% found, almost no false calls.
- Indels are harder (small numbers here: ~30 per sample).
- Main false calls: low-VAF indels in homopolymers; the LowVAF label removes most of them.
- Repeat this benchmark yourself: `scripts/afla-benchmark.sh` (deck exercise 2).



Germline pitfalls to remember

- No coverage, no conclusion: check low-coverage regions before calling a result negative.
- Low VAF "heterozygous" calls are usually artefacts; confirm anything important by Sanger.
- Pseudogenes and repeats (PMS2, SMN1, CYP2D6, repeat expansions) are not reliable with short reads.
- ClinVar changes: re-check with the live button; frequencies matter as much as predictions.
- Suggestions are a start: the classification is yours, with literature and family data.